# Vincent Mallet

Tenure-track Researcher · Ecole des Mines

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## Curriculum

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|---|---|
| 2026 - | **Faculty researcher**, CBIO - Mines Paris |
| 2024 - 2025 | **Tenure-track researcher**, CBIO - Mines Paris |
| 2023 - 2024 | **Postdoctoral researcher Supervised by Maks Ovsjanikov**, Ecole Polytechnique |
| 2022 | **Visiting Student with Emanuele Rodolà**, La Sapienza |
| 2019 - 2022 | **PhD supervised by Michael Nilges and Jean-Philippe Vert**, Institut Pasteur / Mines Paris |
| 2018 - 2019 | **MSc in Computer Science with Jérôme Waldispühl**, McGill Computational Biology Group |
| 2013 - 2019 | **MSc in Bioinformatics**, Ecole Polytechnique |

## Publications

### Journals

**Machine learning for RNA-targeting drug design**. Wissam Karroucha, Carlos Oliver, Veronique Stoven, and Vincent Mallet. *Journal of Chemical Information and Modeling*, 2026. 10.1021/acs.jcim.6c01094

**Structure-based rna drug discovery with deep-learning assisted virtual screening**. Juan Guillermo Carvajal Patiño*, Vincent Mallet*, David Becerra, Luis Fernando Niño, Carlos Oliver, and Jérôme Waldispühl. *Nature Communications*, 2025

**Indeepnet: a web platform for predicting functional binding sites in proteins using indeep**. Fabien Mareuil, Rachel Torchet, Luis Checa Ruano, Vincent Mallet, Michael Nilges, Guillaume Bouvier, and Olivier Sperandio. *Nucleic Acids Research*, page gkaf403, 2025

**Finding antibodies in cryo-em maps with crai**. Vincent Mallet, Chiara Rapisarda, Hervé Minoux, and Maks Ovsjanikov. *Bioinformatics*, 41(5):btaf157, 2025

**Alignscape, displaying sequence similarity using self-organizing maps**. Isaac Filella-Merce, Vincent Mallet, Eric Durand, Michael Nilges, Guillaume Bouvier, and Riccardo Pellarin. *Frontiers in Bioinformatics*, 4, 2024

**InDeep: 3D fully convolutional neural networks to assist in silico drug design on protein-protein interactions**. Vincent Mallet, Luis Checa Ruano, Alexandra Moine Franel, Michael Nilges, Karen Druart, Guillaume Bouvier, and Olivier Sperandio. *Bioinformatics*, 2022

**RNAglib: A python package for RNA 2.5D graphs**. Vincent Mallet, Carlos Oliver, Jonathan Broadbent, William L Hamilton, and Jérôme Waldispühl. *Bioinformatics*, 2022

**VeRNAl: A Tool for Mining Fuzzy Network Motifs in RNA**. Carlos Oliver*, Vincent Mallet*, Pericles Philippopoulos, William L Hamilton, and Jérôme Waldispühl. *Bioinformatics*, 2022

**quicksom: Self-Organizing Maps on GPUs for clustering of molecular dynamics trajectories**. Vincent Mallet, Michael Nilges, and Guillaume Bouvier. *Bioinformatics*, 2021

**Optimol: Optimization of binding affinities in chemical space for drug discovery**. Jacques Boitreaud*, Vincent Mallet*, Carlos Oliver, and Jérome Waldispuhl. *Journal of Chemical Information and Modeling*, 2020

**Augmented base pairing networks encode RNA-small molecule binding preferences**. Carlos Oliver, Vincent Mallet, Roman Sarrazin Gendron, Vladimir Reinharz, William L Hamilton, Nicolas Moitessier, and Jérome Waldispuhl. *Nucleic Acids Research*, 2020

### Conferences Proceedings

**Atomsurf : Surface representation for learning on protein structures**. Vincent Mallet*, Souhaib Attaiki*, Yangyang Miao*, Bruno Correia, and Maks Ovsjanikov. *ICLR*, 2025

**Reverse-complement equivariant networks for dna sequences**. Vincent Mallet and Jean-Philippe Vert. *Advances in Neural Information Processing Systems*, 2021

### Workshops and preprints

**Nicolas Portal, Wissam Karroucha, Vincent Mallet, and Massimiliano Bonomi**. Learning dynamic protein representations at scale with distograms. In *ICLR Workshop on Learning Meaningful Representations of Life (LMRL)*, 2026

**Victor Gertner, Frederic Cazals, and Vincent Mallet**. Alphasurf: On-the-fly surface computations for protein representation learning. In *ICLR Workshop on Learning Meaningful Representations of Life (LMRL)*, 2026

**Luis Wyss, Vincent Mallet, Wissam Karroucha, Karsten Borgwardt, and Carlos Oliver**. A comprehensive library for rna structure-function modeling. In *ICLR Workshop on AI for Nucleic Acids*, 2025

## Awards, Fellowships, & Grants

2025 **Prairie-PSAI Junior Chair**, 150k€
2025 **PhD Fellowship for Victor Gertner**, PSL Graduate Program
2024 **3 PhD Fellowships for Wissam Karroucha**, FRM, PSL, ITN Mines
2022 **Elise Mobility Award**, ELLIS
2019 **2 PhD Fellowship**, INCEPTION Program, AMX

## Presentations

2026. *Machine Learning for RNA structure-based drug design*. Frontier in Biological Science seminar, invited by Zhi Lu and Qiangfeng Zhang, Tsingua University, Beijing, China.

2026. *Machine Learning for RNA structure-based drug design*. Invited talk in Shuangjia Zheng's lab, SJTU Shanghai, China.

2026. *Learning representations for structural biology*. Invited talk in Michael Gromiha's lab, IIT Chennai, India.

2026. *Structure-based drug design and machine learning*. IASD Seminar.

2026. *Machine Learning for RNA structure-based drug design*. Keynote at "AI for the Process InDuStry" (PDS 26) Online.

2026. *RNA Drug Design*. Invited talk at Sanofi Paris.

2026. *Equivariant Neural Networks*. Invited talk in Slavica Jonic's lab, Sorbonne Paris.

2025. *Rationalizing machine learning on biomolecules structures*. Keynote at "LEGO" meeting, Nantes.

2025. *Machine learning on the structure of biomolecules*. talk at "GT Masim", Paris.

2025. *Machine learning on the structure of biomolecules*. Invited Seminar CBS Montpellier.

2024. *Learning on protein structures*. Speaker at "Bits in Bio" Paris.

2024. *Learning on protein structures, with applications in Cryo-EM*. Keynote at "Pasteur Symposium on AI".

2023. *Modeling structural biology with geometric deep learning*. Invited speaker at "Data Shapes Seminar", INRIA Paris.

2023. *Geometric learning for drug design*. Invited speaker at "AstraZeneca drug discovery Seminar" (online).

2023. *RNAGlib: a platform for learning on RNA structure*. Invited speaker at "MILA drug discovery Seminar".

2023. *Reverse-Complement Equivariant Networks for DNA Sequences*. Speaker at "AI in Structural Biology Workshop", Marseille.

2022. *Representation learning on the structure of biomolecules* Invited talk at "GT MASIM", Paris

2022. *InDeep: learning for drug design on protein protein interactions*. Workshop on AI in magnetic resonance, Online.

2022. *Artificial Intelligence in prediction, determination and analysis of structures*. EMBO workshop: Machine learning for structural biology.

## Teaching Experience

2025-2026 **Lecturer**, *Institut Pasteur* Introduction to deep neural networks
2025-2026 **Lecturer**, *Dassault Systèmes* Foundations of deep learning
2024-2026 **Lecturer**, *Ecole des Mines* Modern drug design with learning approaches
2022-2024 **Teacher**, *CRI Paris* 45 hours of classes and practical sessions on Foundations of AI
2022 **Lecturer**, *McGill CS Master* Geometric learning for drug design
2020-2022 **Lecturer**, *Université de Paris, Master ISDD* Generative models for drug design

## Mentoring

### Supervision

2026 **Sioban Nieradzik-Kozik**, M1 Student
2026- **Alexandre Ducorroy**, PhD student
2025- **Victor Gertner**, PhD Student
2025- **Nicolas Portal**, Post-Doc
2025- **Ikram Mahmoudi**, Post-Doc
2024- **Wissam Karroucha**, PhD student
2024- **Victor Laigle**, PhD student

### Joint Mentorship

2026- **Martynas Jakutis**, PhD student, *with Justas Dapkunas, Vilnius University*
2026- **Grisell Diaz**, EMBL Arise Fellow, *with Sameer Velankar, EBI*
2025- **Loïc Chadoutaud**, PhD student, *with Thomas Walter, Institut Curie*
2024- **Mrunali Manjrekar**, PhD student, *with Maz Abulnaga, MIT*
2024- **Yangyang Miao**, PhD student, *with Bruno Correia, EPFL*
2024- **Tamara Künzle**, PhD student, *with Maks Ovsjanikov, Ecole Polytechnique*
2022 **Jacques Boitreaud**, Msc student, *with Jérôme Waldispühl, McGill*

## Additional responsibilities

*Reviewer*: Bioinformatics, NAR, Journal of Chemoinformatics, NeurIPS, ICML, Recomb, ECCB...
*Associate Editor*: Frontiers in Bioinformatics
*Organizer*: GT Masim (French Bioinformatics Community), AlgoSB Winter school
*Member*: Ellis network, PRAIRIE Springboard Chair
*Funding Committee member*: *ANR TSIA Biologie et Santé* 2025 Call
*Doctoral Committee member*: Arthur Carre, François Thénier, Enzo Marsot, Estelle Bergiron, Francesca Venturi, Mehyar Mlaweh, Luca Vitoriano
*Packages*: `rnaglib`, `quicksom`, `CrAI`, `RNAmigos2`